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LAB TASK 4

BECE406E

FPGA System Design Lab

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SLOT	L33+L34
FACULTY NAME	AARTHY M

AIM:

i) Display the time in the format Hr: Min: Sec using clock divider concept and display the output using seven segment display. Attach screenshots of code and FPGA board implementation.

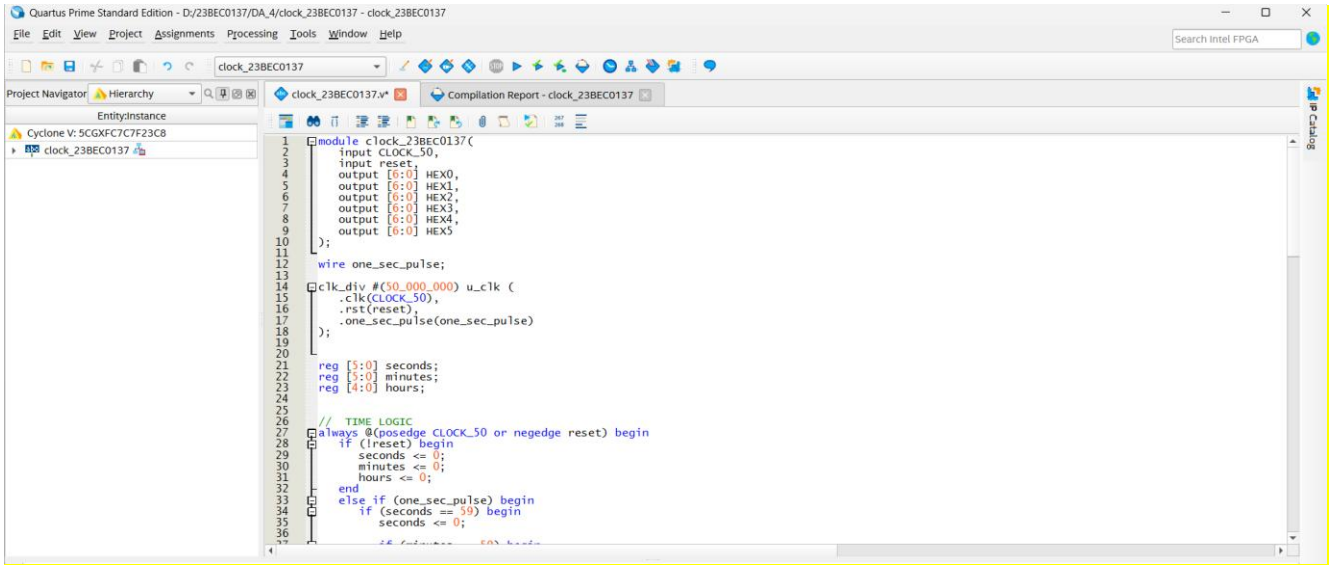
ii) Implement Synchronous FIFO using Quartus and verify the functionality. Verify the timing analysis and power analysis. Check for the set and hold violation and find the critical path. Attach the RTL view of FIFO and Screenshots of code and outputs.

SOFTWARE DETAILS:

Functional Simulation: Modelsim
Synthesis & Implementation: Intel Quartus Prime 23.1

CLOCK

RTL CODE:




```
always @(posedge CLOCK_50 or negedge reset) begin
    if (!reset) begin
        seconds <= 0;
        minutes <= 0;
        hours <= 0;
    end
    else if (one_sec_pulse) begin
        if (seconds == 59) begin
            seconds <= 0;

            if (minutes == 59) begin
                minutes <= 0;

                if (hours == 23)
                    hours <= 0;
                else
                    hours <= hours + 1;
            end
            else
                minutes <= minutes + 1;
        end
        else
            seconds <= seconds + 1;
    end
end

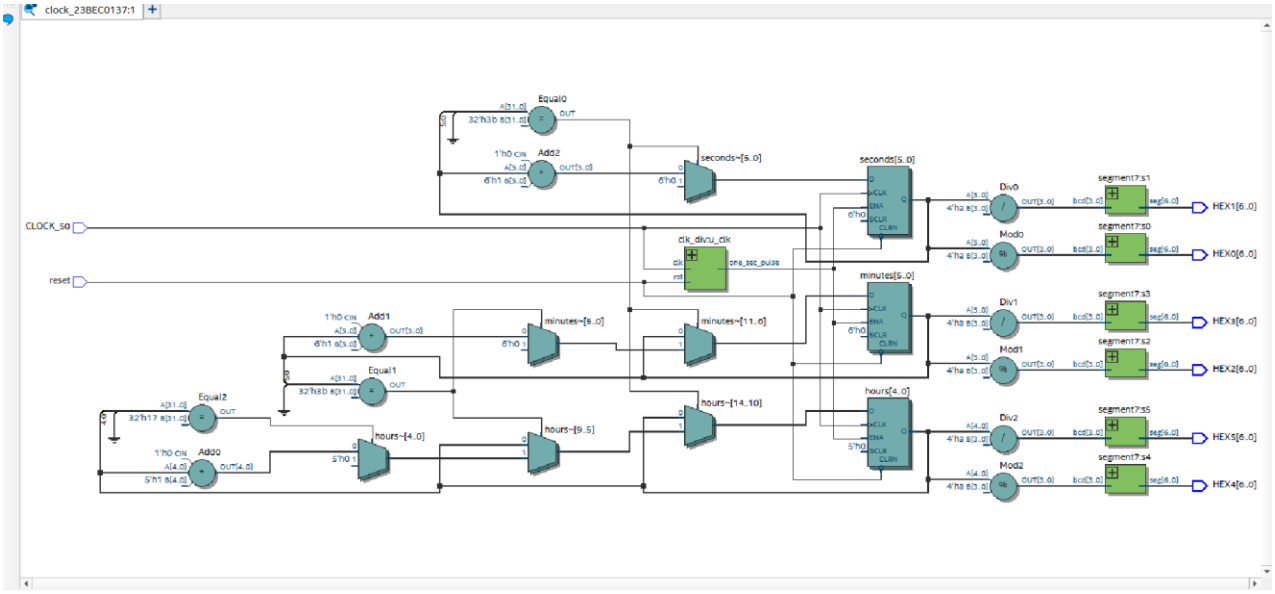
wire [3:0] seconds_unit = seconds % 10;
wire [3:0] seconds_tens = seconds / 10;

wire [3:0] minutes_unit = minutes % 10;
```

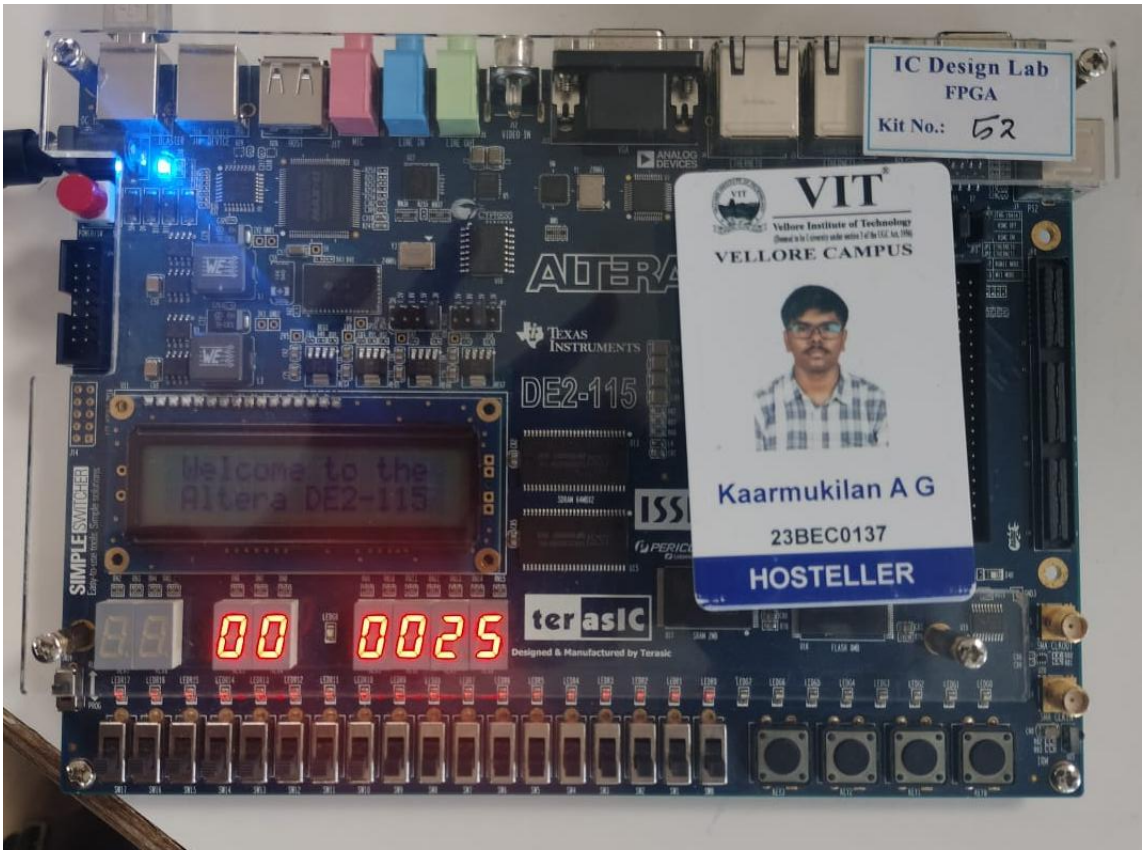
```
52
53
54    wire [3:0] seconds_unit = seconds % 10;
55    wire [3:0] seconds_tens = seconds / 10;
56
57    wire [3:0] minutes_unit = minutes % 10;
58    wire [3:0] minutes_tens = minutes / 10;
59
60    wire [3:0] hours_unit = hours % 10;
61    wire [3:0] hours_tens = hours / 10;
62
63    segment7 s0(seconds_unit, HEX0);
64    segment7 s1(seconds_tens, HEX1);
65
66    segment7 s2(minutes_unit, HEX2);
67    segment7 s3(minutes_tens, HEX3);
68
69    segment7 s4(hours_unit, HEX4);
70    segment7 s5(hours_tens, HEX5);
71
72    endmodule
73
74
75
76    //  PARAMETERIZED DIVIDER
77    module clk_div #(parameter DIV = 50_000_000)(
78        input clk,
79        input rst,
80        output reg one_sec_pulse
81    );
82
83    reg [31:0] count;
84
85    always @(posedge clk or negedge rst) begin
86        if (!rst) begin
87            count <= 0;
88            one_sec_pulse <= 0;
89        end
90        else if (count == DIV - 1) begin
91            count <= 0;
92            one_sec_pulse <= 1;    // 1-cycle pulse
93        end
94        else begin
95            count <= count + 1;
96            one_sec_pulse <= 0;
97        end
98    end
```

```
104   // 7 SEG
105   module segment7(
106     input [3:0] bcd,
107     output reg [6:0] seg
108   );
109
110   always @(*) begin
111     case (bcd)
112         4'd0: seg = 7'b1000000;
113         4'd1: seg = 7'b1111001;
114         4'd2: seg = 7'b0100100;
115         4'd3: seg = 7'b0110000;
116         4'd4: seg = 7'b0011001;
117         4'd5: seg = 7'b0010010;
118         4'd6: seg = 7'b0000010;
119         4'd7: seg = 7'b1111000;
120         4'd8: seg = 7'b0000000;
121         4'd9: seg = 7'b0010000;
122         default: seg = 7'b1111111;
123     endcase
124   end
125
126   endmodule
```

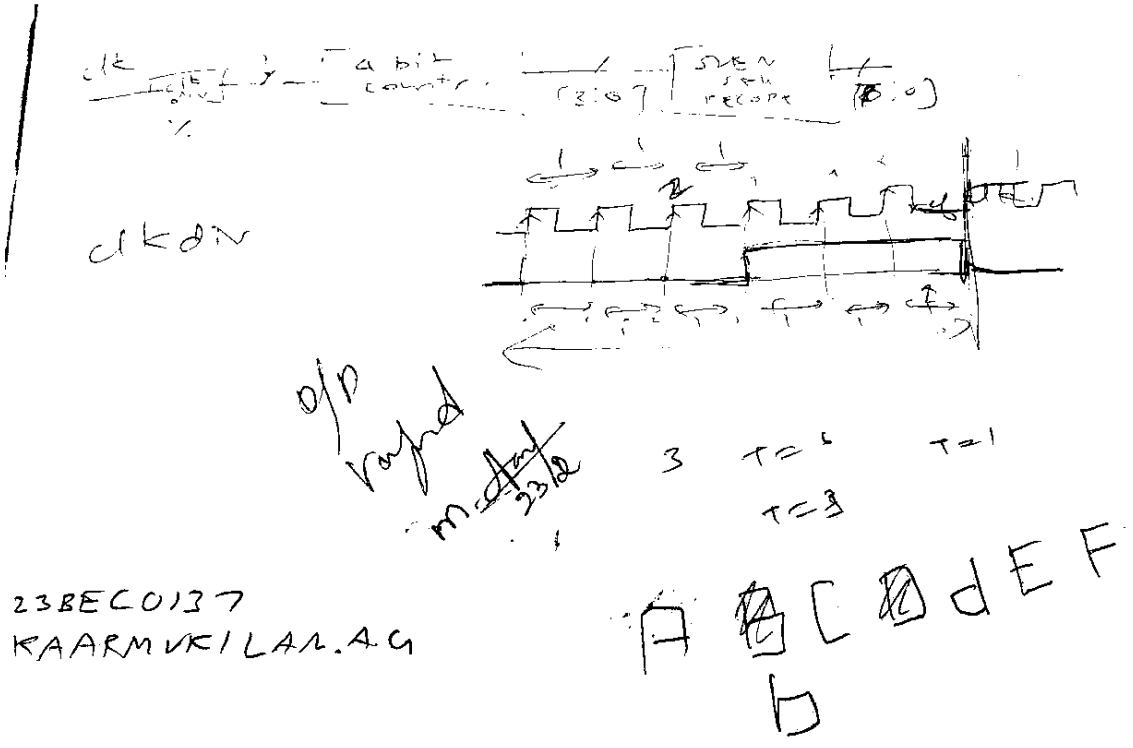
RTL VIEW:



FPGA IMPLEMENTATION:

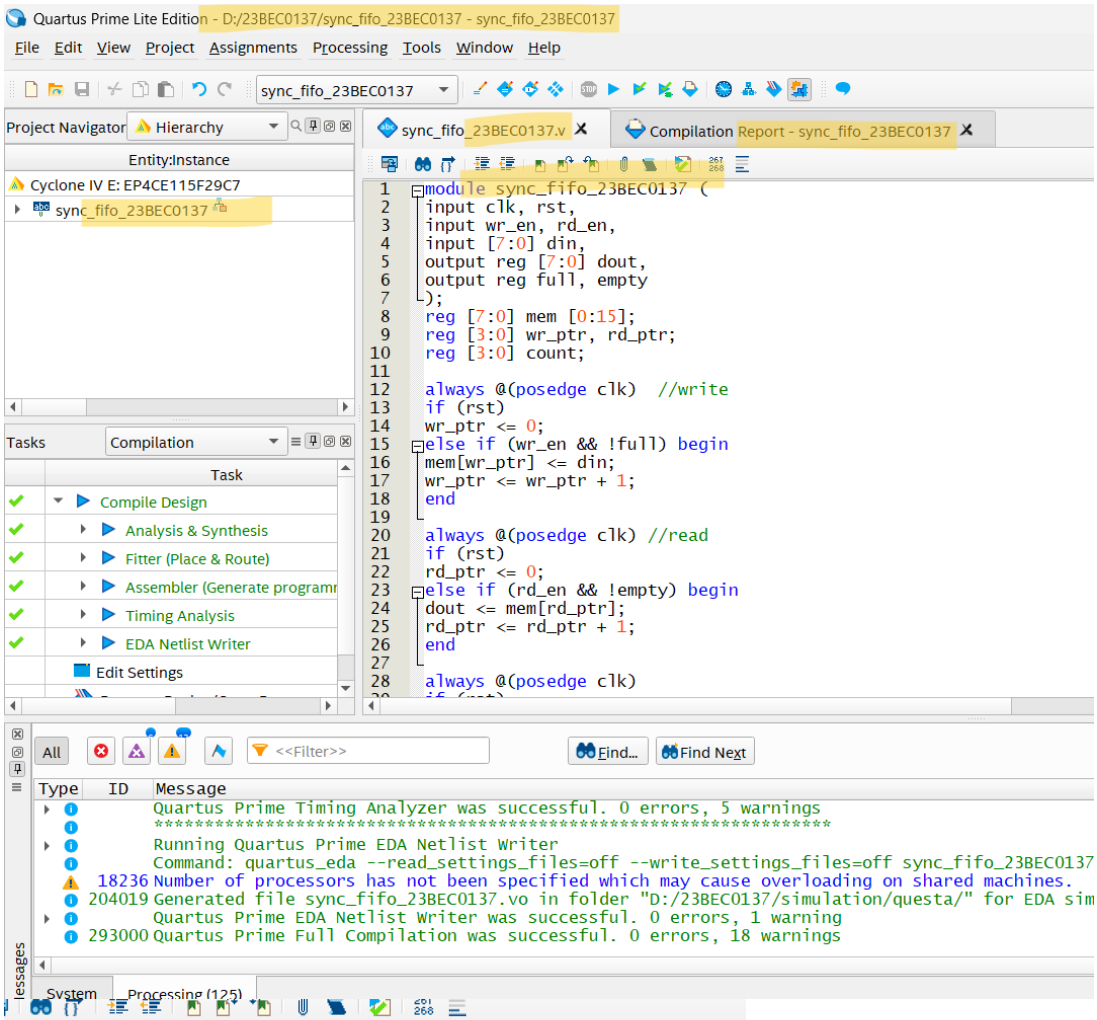


ATTESTATION:



SYNCHRONOUS FIFO

RTL CODE :




```
rd_ptr <= 0;
else if (rd_en && !empty) begin
dout <= mem[rd_ptr];
rd_ptr <= rd_ptr + 1;
end

always @(posedge clk)
if (rst)
count <= 0;
else begin
case ({wr_en && !full, rd_en && !empty})
2'b10: count <= count + 1;
2'b01: count <= count  - 1;
endcase
end

always @(posedge clk) begin
if (rst) begin
empty <= 1;
full  <= 0;
end else begin
empty <= (count == 0);
full  <= (count == 15);
end
end

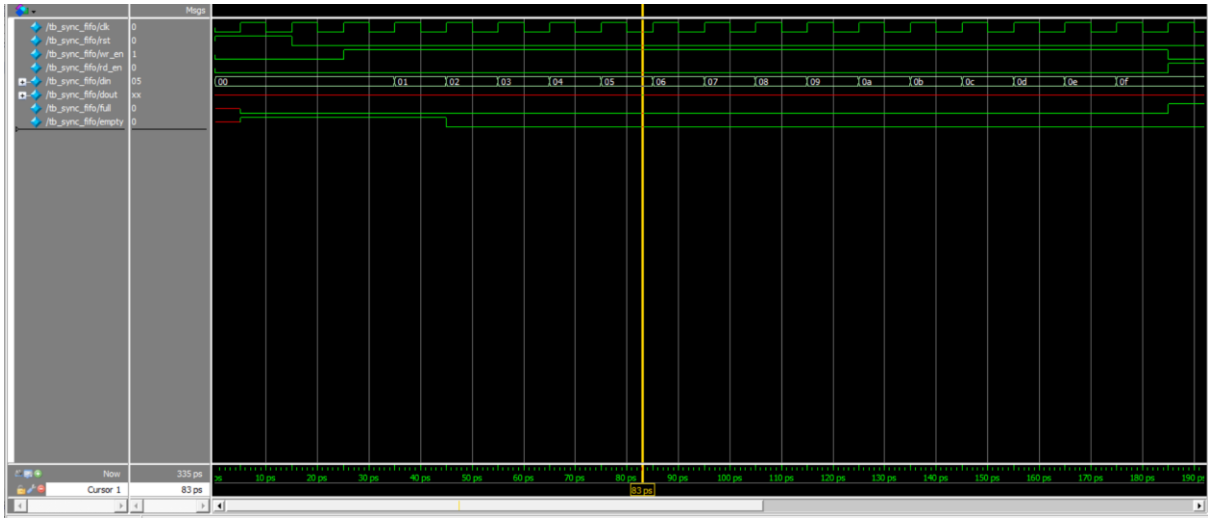
endmodule
```

TESTBENCH:

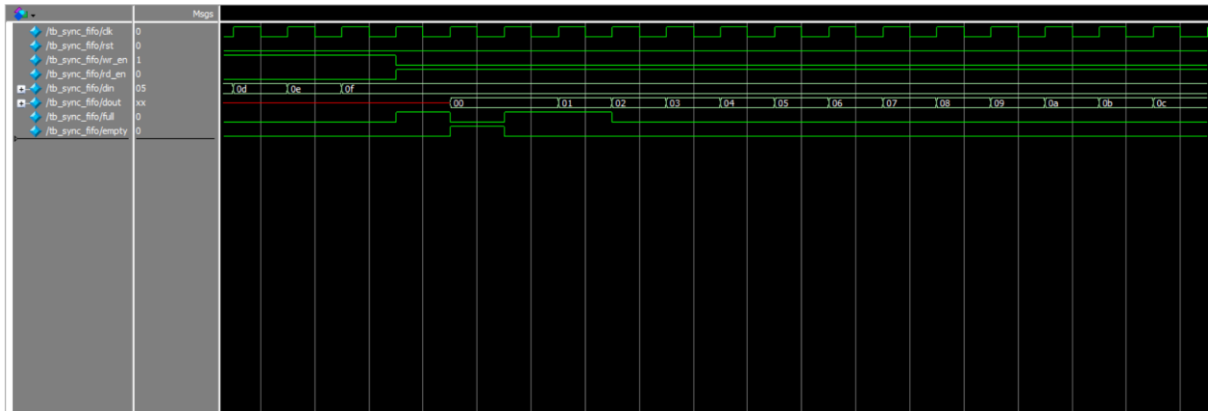
```
Ln#
 1  module tb_sync_fifo_23bec0137;
 2  reg clk, rst;
 3  reg wr_en, rd_en;
 4  reg [7:0] din;
 5  wire [7:0] dout;
 6  wire full, empty;
 7  sync_fifo_23bec0137 dut (
 8      .clk(clk),
 9      .rst(rst),
10      .wr_en(wr_en),
11      .rd_en(rd_en),
12      .din(din),
13      .dout(dout),
14      .full(full),
15      .empty(empty)
16  );
17  always #5 clk = ~clk;
18  integer i;
19  initial begin
20      clk = 0;
21      rst = 1;
22      wr_en = 0;
23      rd_en = 0;
24      din = 0;
25      // Apply reset
26      repeat(2) @(posedge clk);
27      rst = 0;
28      // 1. WRITE UNTIL FULL
29      for (i = 0; i < 16; i = i + 1) begin
30          @(posedge clk);
31          wr_en = 1;
32          rd_en = 0;
33          din = i;
34      end
35      // 2. READ UNTIL EMPTY
36      for (i = 0; i < 16; i = i + 1) begin
37          @(posedge clk);
38          wr_en = 0;
39          rd_en = 1;
40      end
41      $finish;
42  end
43  endmodule
```

FUNCTIONAL SIMULATION:

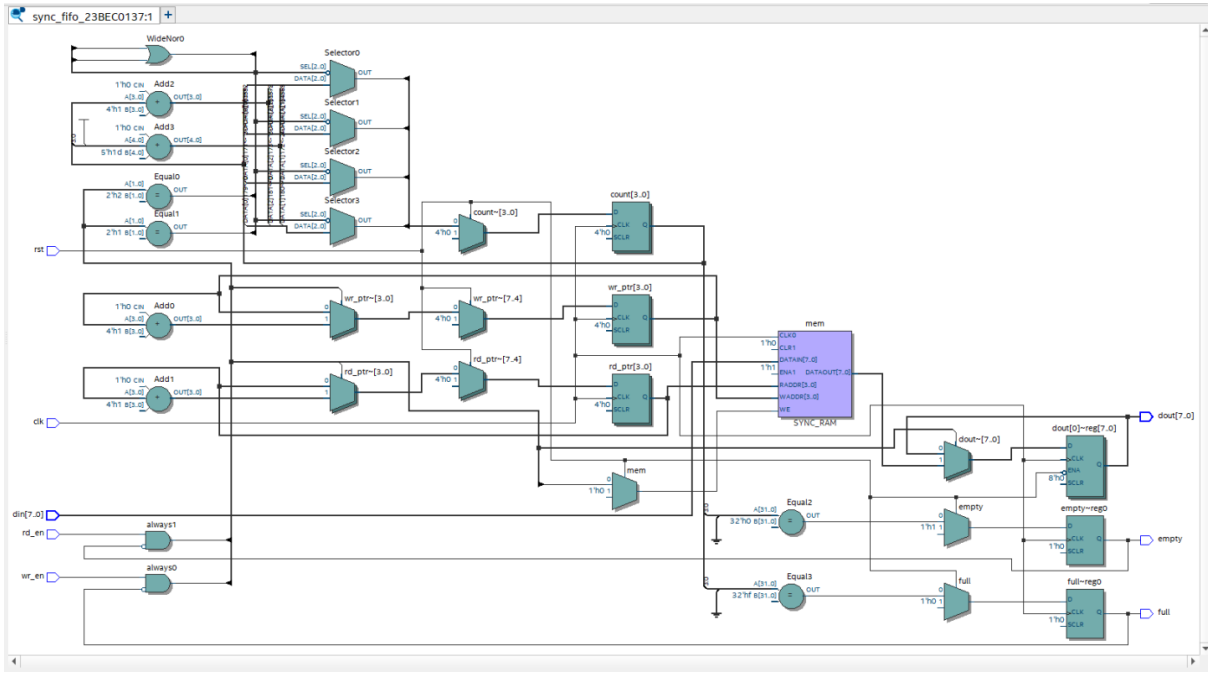
//WRITING 16 LOCATIONS



//READING

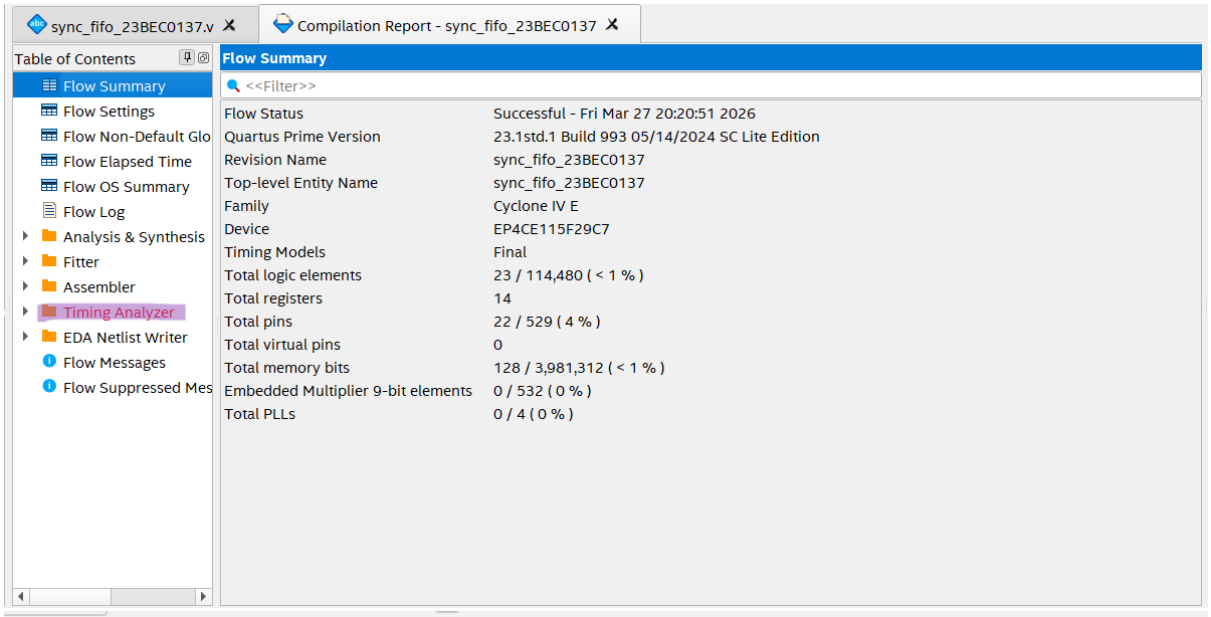


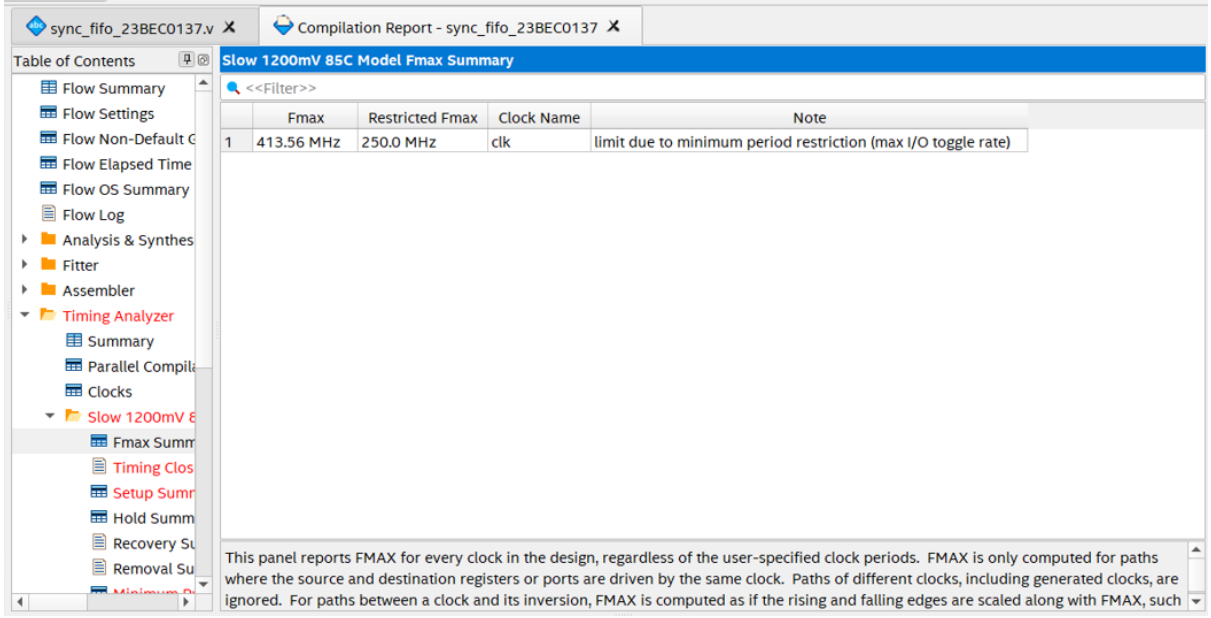
RTL SCHEMATIC:

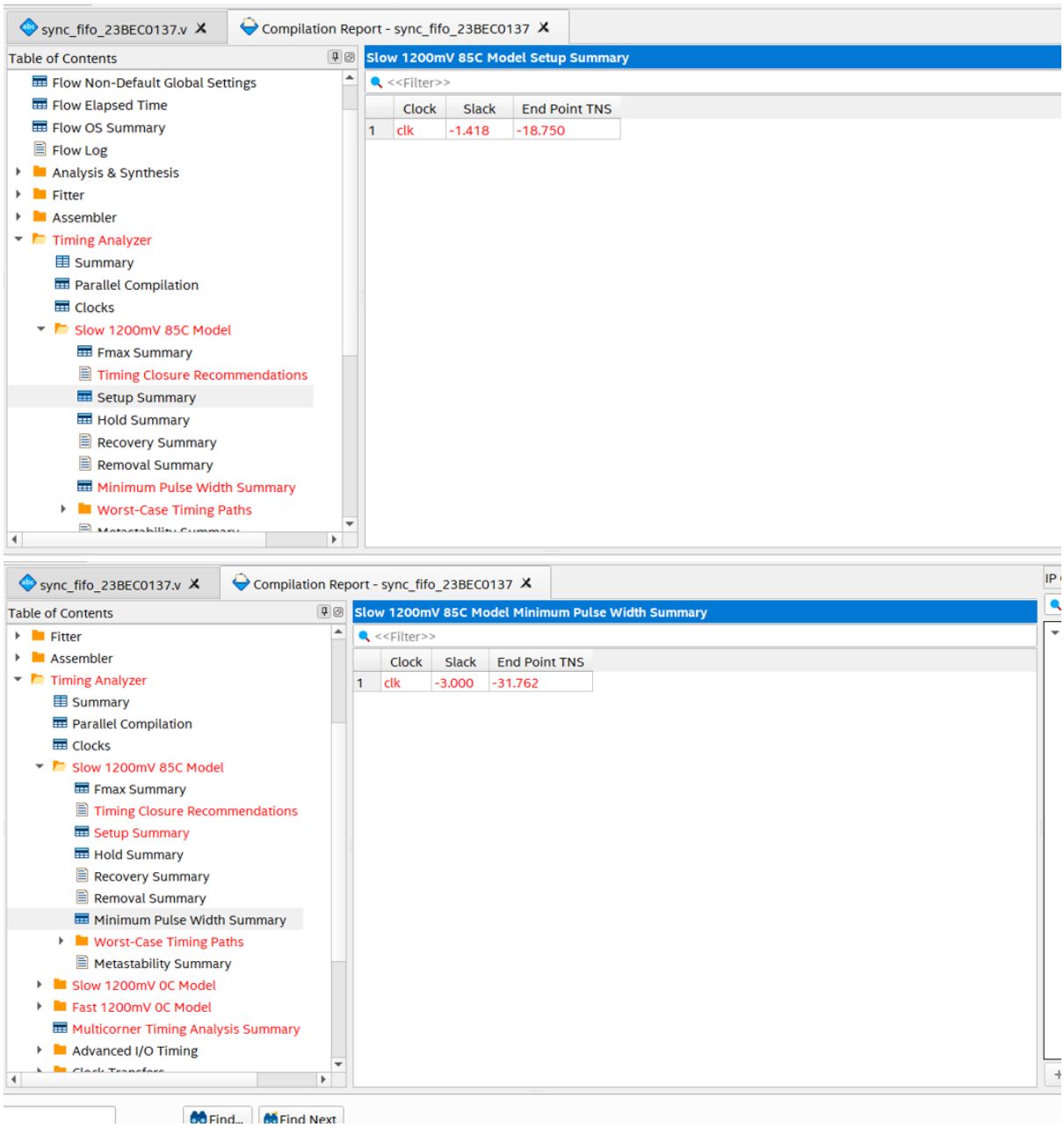


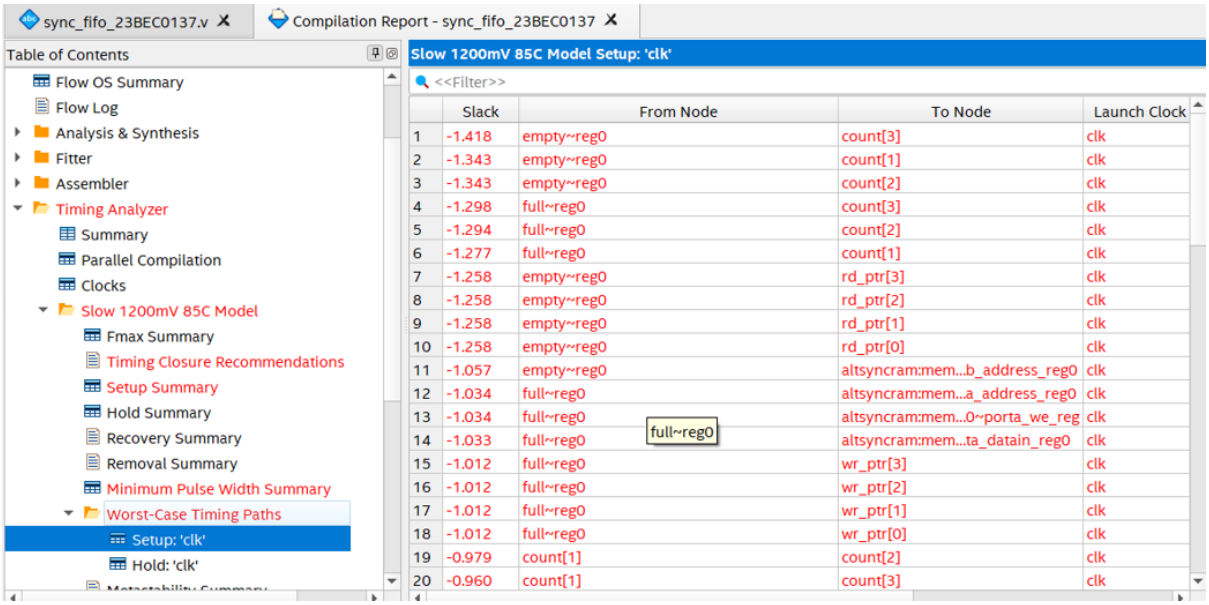
TIMING AND POWER ANALYSIS

BEFORE SETTING SDC:

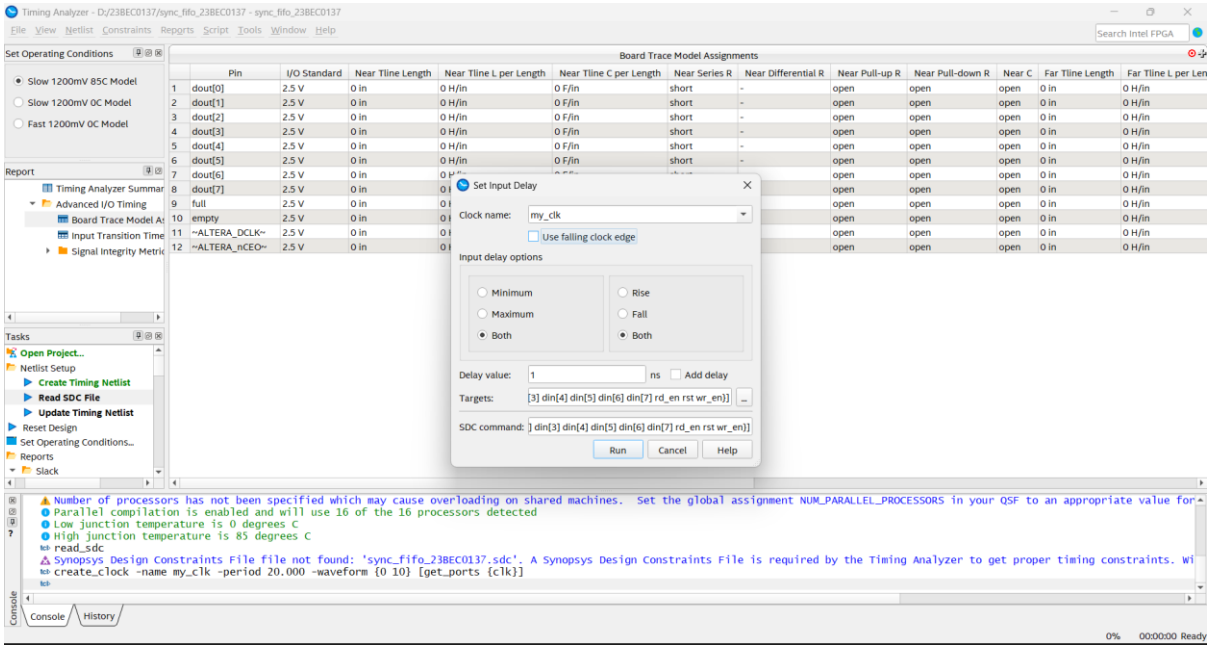


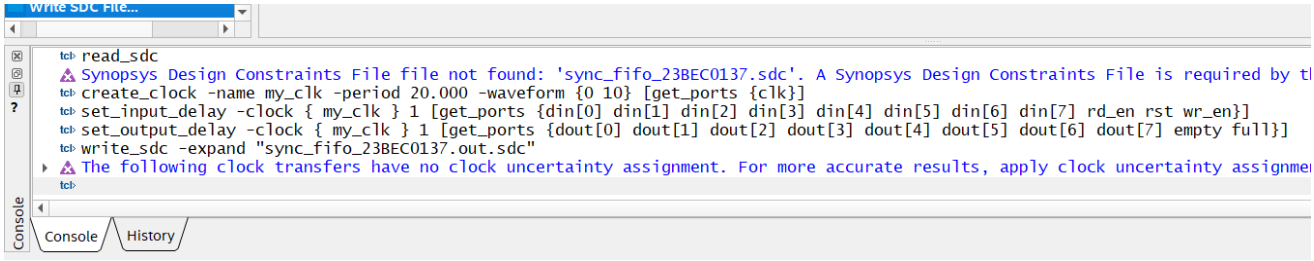


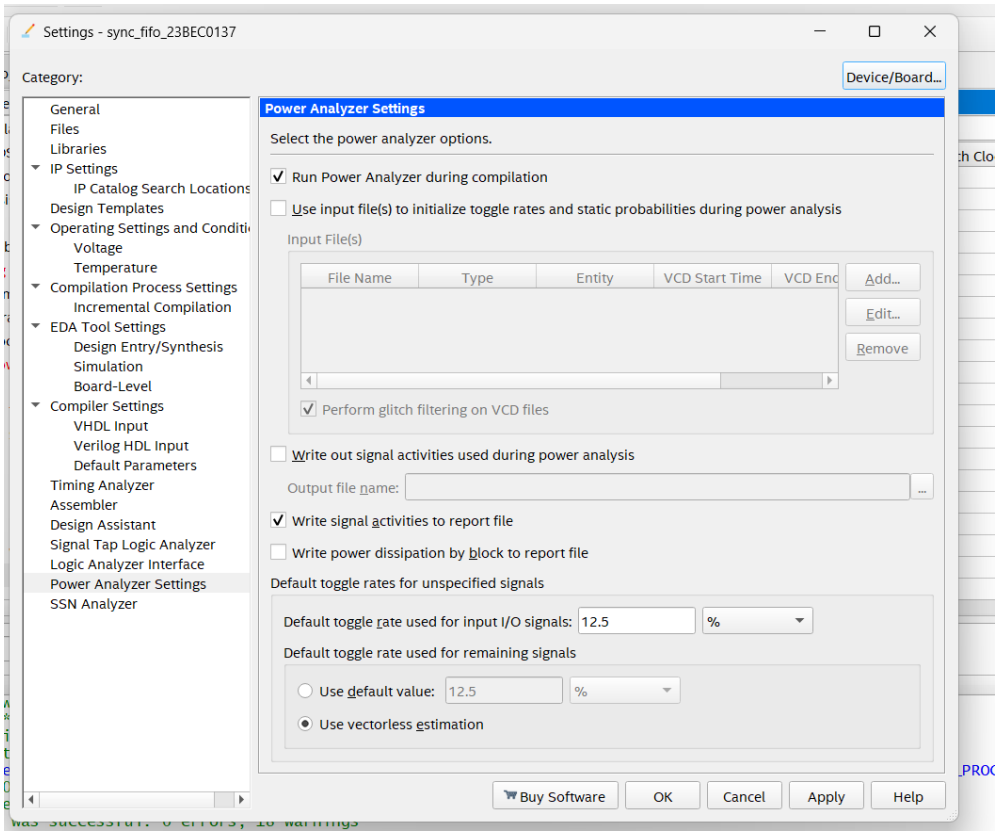

sync_fifo_23BEC0137.v
Compilation Report - sync_fifo_23BEC0137
Table of Contents
Flow Non-Default Global Settings
Flow Elapsed Time
Flow OS Summary
Flow Log
Analysis & Synthesis
Fitter
Assembler
Timing Analyzer
Summary
Parallel Compilation
Clocks
Slow 1200mV 85C Model
Fmax Summary
Timing Closure Recommendations
Setup Summary
Hold Summary
Recovery Summary
Removal Summary
Minimum Pulse Width Summary
Worst-Case Timing Paths
Slow 1200mV 85C Model Setup Summary
<<Filter>>
Clock
Slack
End Point TNS
1
clk
-1.418
-18.750
sync_fifo_23BEC0137.v
Compilation Report - sync_fifo_23BEC0137
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Fitter
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Fmax Summary
Timing Closure Recommendations
Setup Summary
Hold Summary
Recovery Summary
Removal Summary
Minimum Pulse Width Summary
Worst-Case Timing Paths
Metastability Summary
Slow 1200mV 0C Model
Fast 1200mV 0C Model
Multicorner Timing Analysis Summary
Advanced I/O Timing
Slow 1200mV 85C Model Minimum Pulse Width Summary
<<Filter>>
Clock
Slack
End Point TNS
1
clk
-3.000
-31.762
Find...
Find Next


sync_fifo_23BEC0137.v
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Table of Contents
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Setup Summary
Hold Summary
Recovery Summary
Removal Summary
Minimum Pulse Width Summary
Worst-Case Timing Paths
Setup: 'clk'
Hold: 'clk'
Slow 1200mV 85C Model Setup: 'clk'
<<Filter>>
Slack
From Node
To Node
Launch Clock
1 -1.418 empty~reg0 count[3] clk
2 -1.343 empty~reg0 count[1] clk
3 -1.343 empty~reg0 count[2] clk
4 -1.298 full~reg0 count[3] clk
5 -1.294 full~reg0 count[2] clk
6 -1.277 full~reg0 count[1] clk
7 -1.258 empty~reg0 rd_ptr[3] clk
8 -1.258 empty~reg0 rd_ptr[2] clk
9 -1.258 empty~reg0 rd_ptr[1] clk
10 -1.258 empty~reg0 rd_ptr[0] clk
11 -1.057 empty~reg0 altsyncram:mem...b_address_reg0 clk
12 -1.034 full~reg0 altsyncram:mem...a_address_reg0 clk
13 -1.034 full~reg0 altsyncram:mem...0~porta_we_reg clk
14 -1.033 full~reg0 altsyncram:mem...ta_datain_reg0 clk
15 -1.012 full~reg0 wr_ptr[3] clk
16 -1.012 full~reg0 wr_ptr[2] clk
17 -1.012 full~reg0 wr_ptr[1] clk
18 -1.012 full~reg0 wr_ptr[0] clk
19 -0.979 count[1] count[2] clk
20 -0.960 count[1] count[3] clk
full~reg0

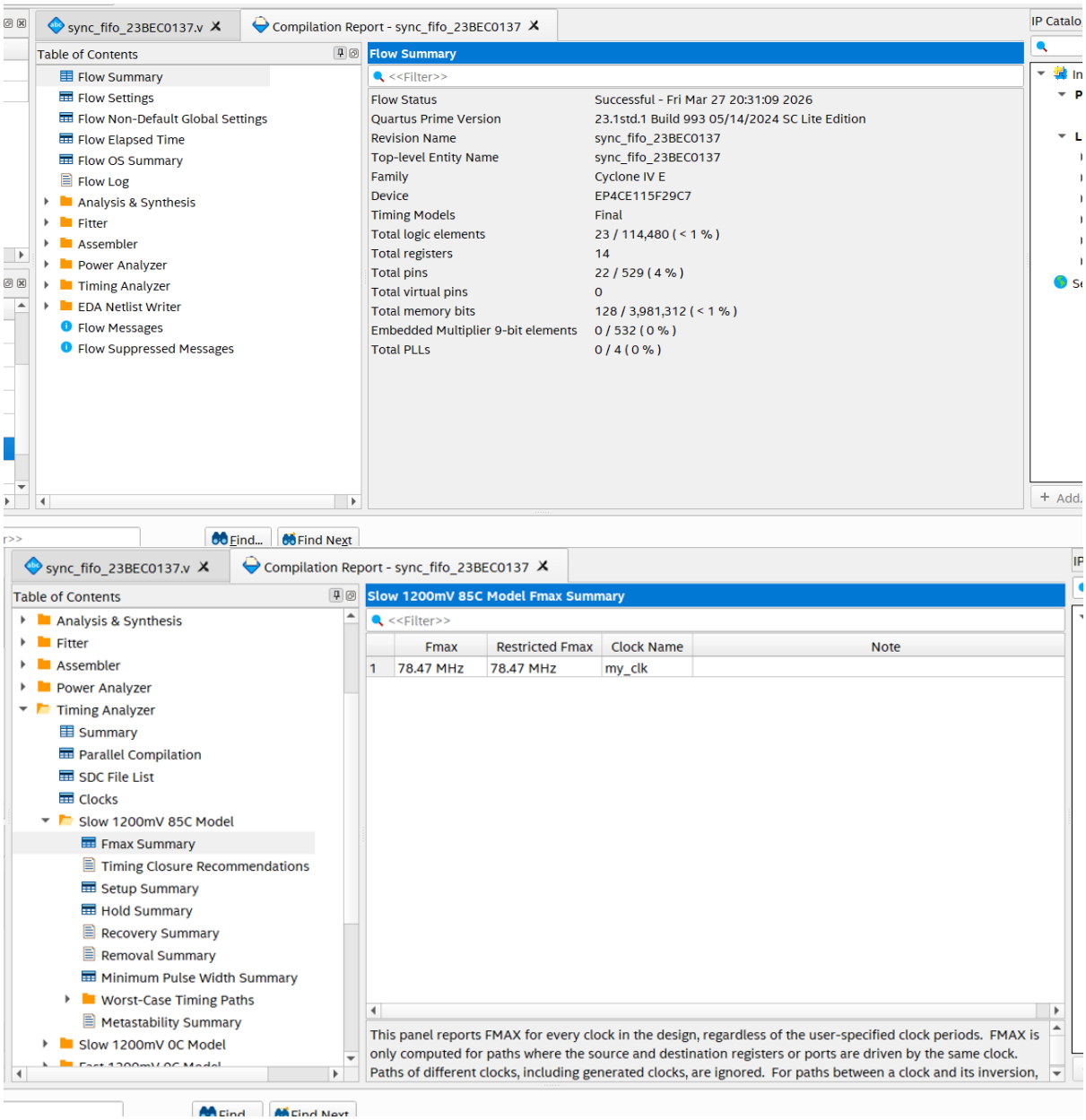
READ AND WRITE SDC IN TIMING ANALYSER TOOL IN QUARTUS PRIME

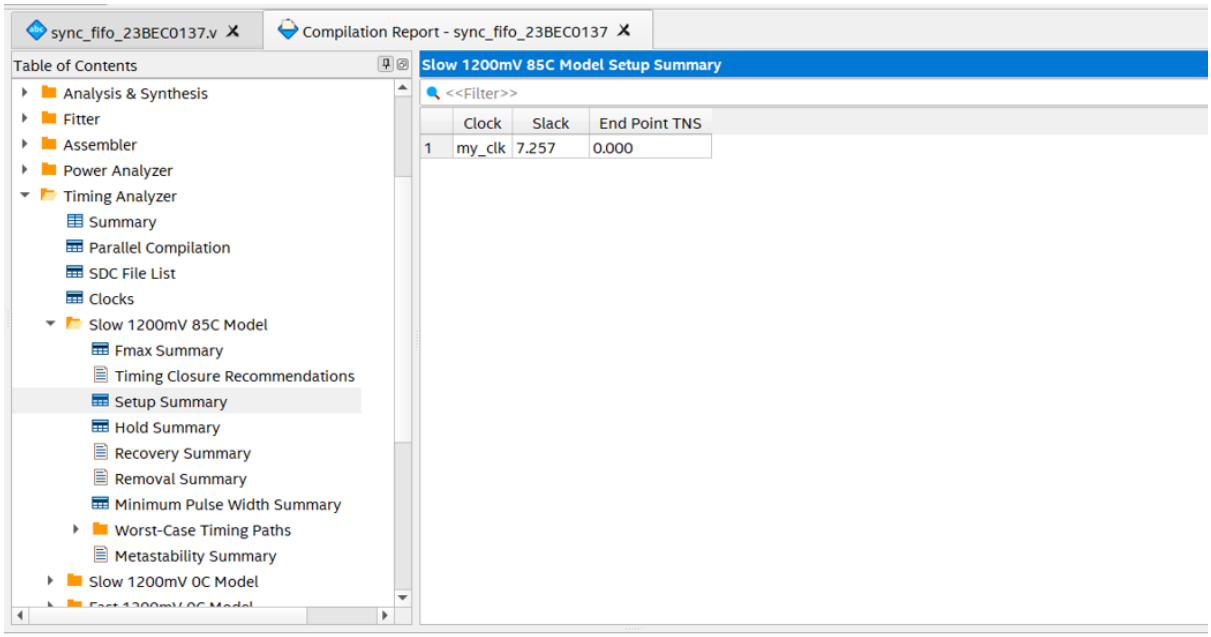


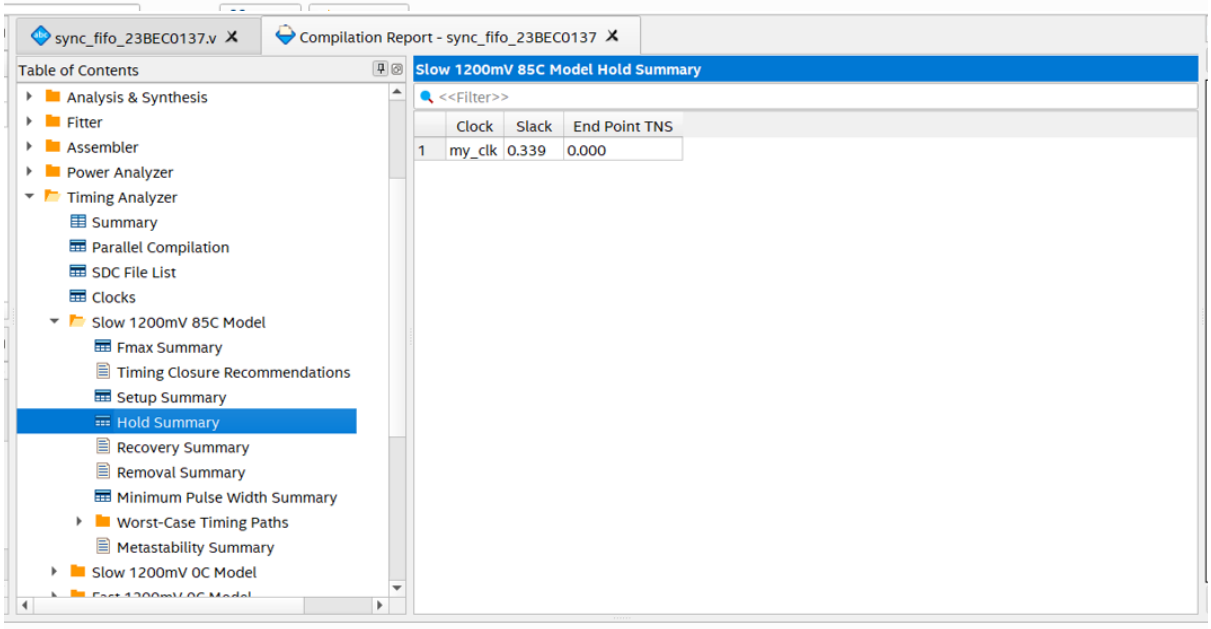




AFTER SETTING SDC FILE:




sync_fifo_23BEC0137.v
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Recovery Summary
Removal Summary
Minimum Pulse Width Summary
Worst-Case Timing Paths
Metastability Summary
Slow 1200mV 0C Model
Slow 1200mV 85C Model Setup Summary
<<Filter>>
Clock
Slack
End Point TNS
1
my_clk
7.257
0.000


sync_fifo_23BEC0137.v
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Removal Summary
Minimum Pulse Width Summary
Worst-Case Timing Paths
Metastability Summary
Slow 1200mV 0C Model
Slow 1200mV 85C Model Hold Summary
<<Filter>>
Clock
Slack
End Point TNS
1
my_clk
0.339
0.000

CRITIICAL PATH – with lowest slack/highest Data delay

sync_fifo_23BEC0137.v | Compilation Report - sync_fifo_23BEC0137

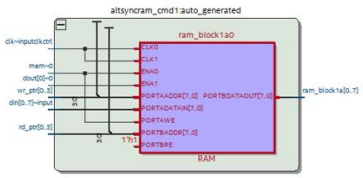
Table of Contents

Slow 1200mV 85C Model Setup: 'my_clk'

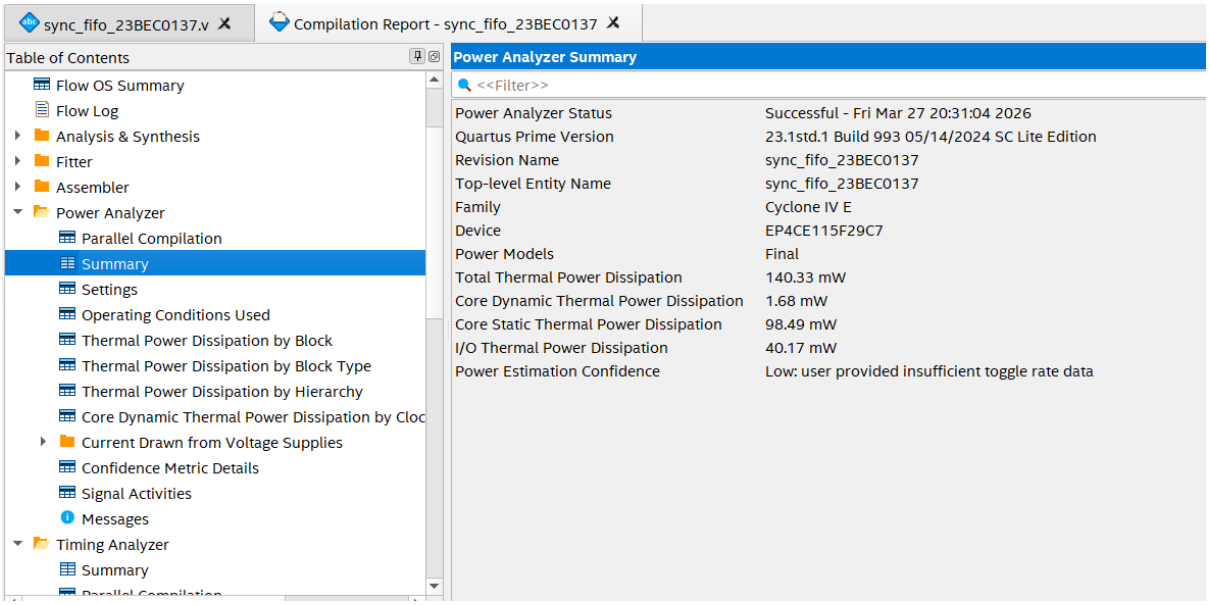
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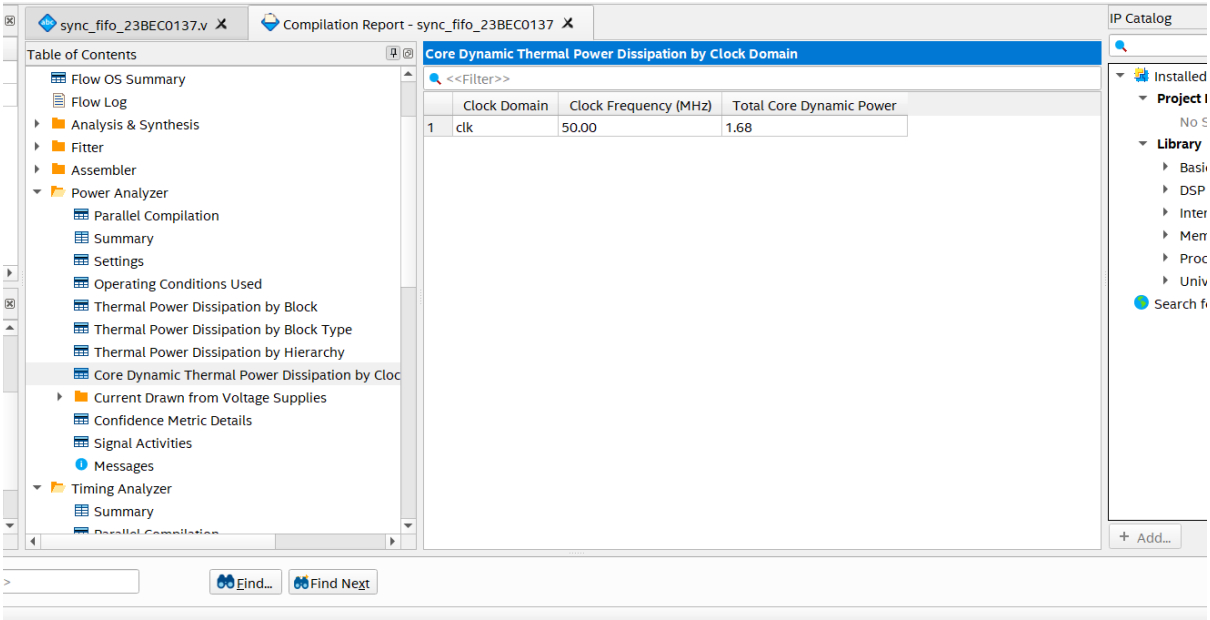
	Slack	From Node	To Node	Launch Clock	Latch Clock	Relationship	Clock Skew	Data Delay
1	7.257	altsyncram:mem_...tb_address_reg0	dout[5]	my_clk	my_clk	20.000	-3.365	8.378
2	8.481	altsyncram:mem_...tb_address_reg0	dout[2]	my_clk	my_clk	20.000	-3.365	7.154
3	8.488	altsyncram:mem_...tb_address_reg0	dout[3]	my_clk	my_clk	20.000	-3.365	7.147
4	8.526	altsyncram:mem_...tb_address_reg0	dout[1]	my_clk	my_clk	20.000	-3.365	7.109
5	8.660	altsyncram:mem_...tb_address_reg0	dout[6]	my_clk	my_clk	20.000	-3.365	6.975
6	8.694	altsyncram:mem_...tb_address_reg0	dout[7]	my_clk	my_clk	20.000	-3.365	6.941
7	8.764	altsyncram:mem_...tb_address_reg0	dout[4]	my_clk	my_clk	20.000	-3.365	6.871
8	8.790	altsyncram:mem_...tb_address_reg0	dout[0]	my_clk	my_clk	20.000	-3.365	6.845
9	11.759	full~reg0	full	my_clk	my_clk	20.000	-3.079	4.162
10	11.942	empty~reg0	empty	my_clk	my_clk	20.000	-3.081	3.977
11	14.900	wr_en	count[3]	my_clk	my_clk	20.000	2.967	7.085
12	15.051	wr_en	count[2]	my_clk	my_clk	20.000	2.967	6.934
13	15.157	rd_en	count[2]	my_clk	my_clk	20.000	2.967	6.828
14	15.176	rd_en	count[3]	my_clk	my_clk	20.000	2.967	6.809
15	15.411	wr_en	count[1]	my_clk	my_clk	20.000	2.967	6.574
16	15.521	rd_en	count[1]	my_clk	my_clk	20.000	2.967	6.464
17	15.652	rst	altsyncram:mem_r...1a0~porta_we_reg	my_clk	my_clk	20.000	3.325	6.731
18	15.652	rst	altsyncram:mem_rt...orta_address_reg0	my_clk	my_clk	20.000	3.325	6.731
19	15.653	rst	altsyncram:mem_rt...porta_datain_reg0	my_clk	my_clk	20.000	3.331	6.736
20	15.700	rst	altsyncram:mem_rt...ortb_address_reg0	my_clk	my_clk	20.000	3.257	6.615
21	15.790	rst	rd_ptr[3]	my_clk	my_clk	20.000	2.966	6.194

sync_fifo_23BEC0137:1



POWER ANALYSER:





INFERENCE:

Hence, STA and power analysis has been done for Synchronous FIFO.

- Maximum operating frequency has been **reduced**, **Negative slack** has been **eliminated** and the **RED lines** in compilation report has **vanished**